METHOD 25 / A COMPLETE BEGINNER GUIDE

Volatility and realized correlation

Two different ways FX co-movement shows up in volatility. Volatility correlation tracks whether absolute moves align. Realized correlation sums intraday return products within a day.

These are **two separate calculations**. Do not mix daily closing prices into the realized formula when your question needs intraday buckets. Teaching numbers only; experiments not started.

Read it in three parts

First: Definitions and the FX percent example. **Next:** Formulas and hand work for rho_vol and rho_realized. **Last:** Hayashi-Yoshida (2005) context, limits, and folder 26 link.

Pages	What you will learn
2	Two tools, not one
3	FX 1.1000 to 1.1110 percent
4	Which series each formula uses
5	Hayashi & Yoshida 2005 context
6-9	Symbols, formulas, two hand calcs
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12-14	Limits, software, sources

Prepared for the folder: 25 - Volatility and Realized Correlation
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02 / THE BASIC IDEA

Two tools, one folder

Volatility correlation

```
rho_vol = Corr( |x_t|, |y_t| )
```

Apply Pearson (folder 01) to absolute returns (or another volatility proxy) on the same bar clock. It asks whether big moves in X tend to coincide with big moves in Y, ignoring sign.

Realized correlation within day d

```
rho_realized,d = SUM_k x(d,k)*y(d,k) /
sqrt( SUM_k x(d,k)^2 * SUM_k y(d,k)^2 )
```

Here x(d,k) and y(d,k) are intraday returns in bucket k on day d. The formula is a correlation shape applied to within-day vectors, not to one daily close pair.

Is either score a price percent?

No. Both live on [-1, +1] (when defined). A 1.00% return is input data, not rho_vol or rho_realized.

03 / RETURNS IN PLAIN NUMBERS

FX percent math on one move

```
Simple: 1.1110/1.1000 - 1 = 0.01 (1.00%)
Log: ln(1.1110/1.1000) = about 0.009950
```

Volatility correlation uses a series of such bar returns, then takes absolute values. Realized correlation splits a day into k buckets (for example 30-minute legs) and uses those bucket returns inside the sums.

Keep one return definition per study. Mixing simple |x| with log y breaks comparability.

04 / DATA RULES

Which data for which formula?

rho_vol

Needs aligned bar returns x_t and y_t on the same schedule. Weekly or daily bars are common. You correlate |x_t| with |y_t|, not raw signed returns unless your question says otherwise.

rho_realized,d

Needs intraday returns x(d,k) and y(d,k) for the same day d. Using only two daily closing prices to stand in for SUM_k x(d,k)y(d,k) is the wrong input for this realized definition.

Async quotes

If buckets do not line up in time, see folder 26 (Hayashi-Yoshida covariance) for overlap-aware covariance. Realized correlation here assumes aligned buckets within the day.

Research context

Hayashi, T., & Yoshida, N. (2005) [1]

Developed covariance and correlation estimators for asynchronously observed processes. Realized FX research often cites this line when closing-time mismatch would bias naive covariances.

This folder separates (a) volatility co-movement rho_vol and (b) within-day realized rho_realized,d. Folder 26 gives the overlap-sum covariance when timestamps differ.

What we claim

Teaching arithmetic only. No invented EURUSD realized paths. When experiments start, log bucket size, timezone, and whether rho_realized is averaged across days.

06 / DICTIONARY

Meet the symbols

Symbol	Meaning
x_t, y_t	Bar returns (often log) at time t.
|x_t|	Absolute return or volatility proxy at t.
rho_vol	Pearson correlation of |x| and |y| series.
x(d,k)	Intraday return in bucket k on day d.
rho_realized,d	Within-day correlation of intraday vectors.

Some papers average rho_realized,d over d to get a series. State your aggregation rule explicitly in reports.

07 / RHO_VOL

Volatility correlation formula

```
rho_vol = Corr( |x_1|,|y_1| ), ..., ( |x_n|,|y_n| )
```

Use the Pearson steps from folder 01 on the absolute-value lists. Zero returns contribute zeros to |x|; they are not missing unless you drop the bar.

Interpretation

High rho_vol means volatility shocks tend to arrive together. Signed return correlation can be low while rho_vol is high if both legs often move a lot in opposite directions.

08 / RHO_REALIZED,D

Realized correlation formula

```
num_d = SUM_k x(d,k) * y(d,k)
den_d = sqrt( SUM_k x(d,k)^2 * SUM_k y(d,k)^2 )
rho_realized,d = num_d / den_d
```

For a single day this is the cosine similarity of the two intraday return vectors. It equals their Pearson correlation when both lists have varying values.

Not the same as rho_vol

rho_vol compares volatility levels across days (or bars). rho_realized,d compares intraday co-movement patterns inside one day. Compute both only when your question needs both.

09 / WORKED EXAMPLE A

Hand calc: rho_vol

Three made-up daily returns (decimals):

t	x_t	y_t	|x_t|	|y_t|
1	0.010	0.012	0.010	0.012
2	-0.005	0.004	0.005	0.004
3	0.008	-0.006	0.008	0.006

```
rho_vol = Pearson(|x|, |y|) = about 0.9226 in this toy list
```

Opposite signed returns on row 3 still have positive absolute values, so volatility can align even when direction differs.

Hand calc: rho_realized,d

One teaching day with three intraday buckets:

k	x(d,k)	y(d,k)
1	0.004	0.005
2	0.003	0.002
3	0.003	0.005

```
num = 0.004*0.005 + 0.003*0.002 + 0.003*0.005 = 0.000041
sum x^2 = 0.000034, sum y^2 = 0.000054
den = sqrt(0.000034 * 0.000054)
rho_realized,d = about 0.9569
```

If you only had daily closes for that day, you would have one return pair, not this three-bucket vector. That is why daily closes are the wrong input for this realized formula.

11 / COMMON ERROR

Daily closes vs intraday buckets

Wrong shortcut

Taking open-to-close once per day and plugging it into SUM_k x(d,k)y(d,k) mislabels a single return as a full intraday sum.

Right habit

Store bucket timestamps, enforce equal bucket rules or document irregular spacing, and compute rho_realized,d per day before any cross-day average.

For volatility correlation, daily closes are fine if x_t and y_t are daily returns and you want rho_vol on daily |x| and |y|.

Limits

Not hedge ratios

Correlation of absolute returns does not give a beta for levels or a cointegration hedge (folder 27).

Microstructure noise

Very fine buckets add noise; very coarse buckets hide within-day structure.

Versus folder 26

Hayashi-Yoshida sums overlapping intervals. Realized rho here assumes aligned intraday buckets.

Software sketch

```
from math import sqrt

def pearson(x, y):
mx, my = sum(x)/len(x), sum(y)/len(y)
dx = [a-mx for a in x]; dy = [b-my for b in y]
t = sum(a*b for a,b in zip(dx,dy))
b = sqrt(sum(a*a for a in dx)*sum(b*b for b in dy))
return t/b

rho_vol = pearson([0.01,0.005,0.008],[0.012,0.004,0.006])
# Realized: intraday lists in INTRA_X, INTRA_Y
```

Libraries for realized volatility often provide bucketed returns; map their output to x(d,k) before you apply the formula on page 8.

Sources and further reading

[1] Hayashi, T., & Yoshida, N. (2005). On covariance estimation of non-synchronously observed diffusion processes. Bernoulli, 11(4), 729-749.
Async covariance foundation; related to realized FX practice.
Open source

[2] Folder 01 Pearson Correlation guide.
Pearson steps used inside rho_vol.
Open source

[3] Folder 26 Hayashi-Yoshida Covariance.
Overlap covariance when buckets are not aligned.
Open source

Next: experiments/v001 with logged bucket grid and separate CSV columns for rho_vol series and daily rho_realized,d.

Checklist before experiments

1. Name which measure you need

Volatility co-movement (rho_vol) versus within-day pattern (rho_realized,d). Do not report one as the other.

2. Build returns consistently

Same quote convention, timezone, and log-or-simple choice. Include the 1.1000 to 1.1110 percent example in notes as a unit check.

3. Validate inputs

For realized work, count buckets per day and flag days with missing intraday data instead of silently reusing closes.

4. Report uncertainty

Toy lists here omit confidence bands. Real studies should state sample length and stability across subsamples.

Experiments not started. No live pair claims belong in this folder until v001 exists with snapshot id.